

AI FDE End-to-End Capability Stack

AI FDE Operating Model mapping

SPECIFY • BUILD • VERIFY • SHIP • REPEAT

Spec-Driven Development with AI Cursor

From problem to production — specifications first, working software with evidence.

CLEAR SPECS
BETTER SOFTWARE
HIGHER CONFIDENCE

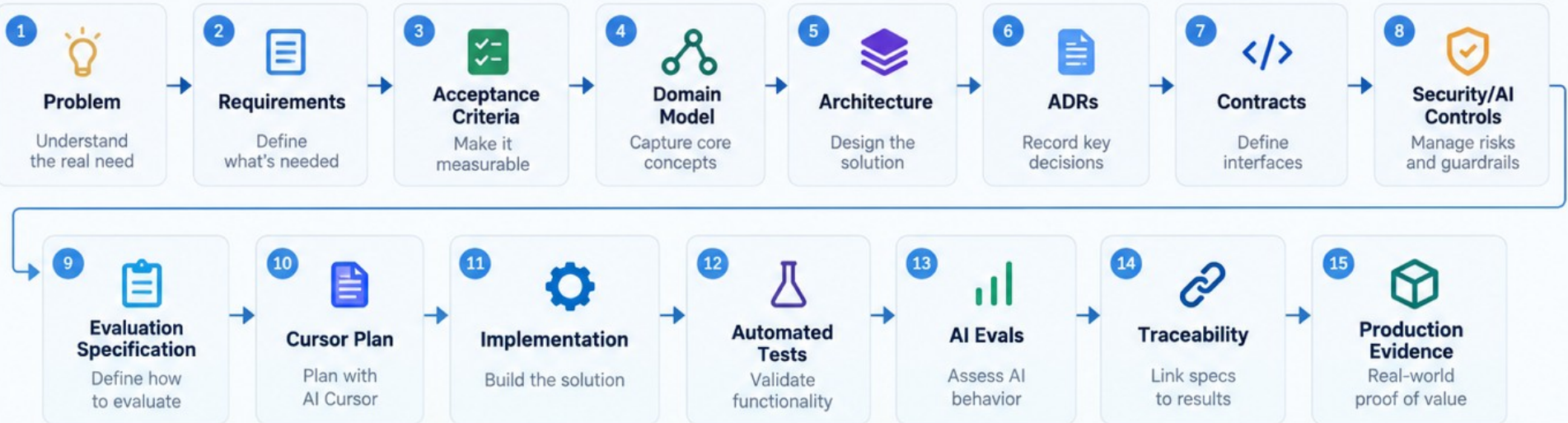


THE DISCIPLINED PATH

A specification-led flow from idea to impact.



SPECS → EXECUTION → EVIDENCE → TRUST



Biggest mistake

The prompt-and-hope approach leads to brittle, risky software.

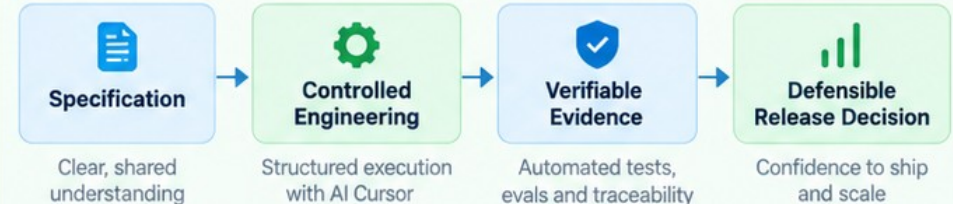


✗ **Fast at first. Expensive later.**



For serious AI FDE work

A disciplined approach delivers higher quality, lower risk, and real value.



✓ **Slower to start. Far greater returns.**

AI FDE Operating Model - phase index

Reference phases used in the mapping below.

Seq.	Workflow	Seq.	Workflow
1	Mandate and field immersion	12	Design security, guardrails and supplier controls
2	Discover process and architecture	13	Approve ADRs and delivery specification
3	Frame problem, root cause and value	14	Engineer
4	Triage regulation and qualify use case	15	Evaluate, attack and independently assure
5	Model the domain	16	Prepare operations, recovery and regulatory evidence
6	Qualify data and knowledge	17	Deploy progressively and integrate adoption
7	Define evaluations, impacts and risks	18	Monitor and validate operational resilience
8	Generate, test and select options	19	Prove value and tell the decision story
9	Design information architecture	20	Evaluate AIMS and decide lifecycle state
10	Design AI and application architecture	21	Retire and capture reusable IP
11	Design agentic and multi-agent orchestration		

AI FDE End-to-End Capability Stack

Mapped to the AI FDE Operating Model

Explicit - named directly in the Operating Model **Supported** - covered by explicit methods, artifacts or controls **Not explicit** - not explicitly named in the Operating Model

Capability	Mapping	Exact Operating Model source anchor(s)
Consulting judgement	Supported	P1 - Mandate and field immersion Approved mandate and operating context P4 - Triage regulation and qualify use case Go/no-go and kill criteria P8 - Generate, test and select options Weighted trade-off matrix P19 - Prove value and tell the decision story Decision recommendation
Stakeholder discovery	Explicit	P1 - Mandate and field immersion Stakeholder discovery
Problem framing and problem qualification	Explicit	P3 - Frame problem, root cause and value SCQA · DMAIC Analyse · Root-cause analysis · Baseline KPIs · Data-driven framing P4 - Triage regulation and qualify use case AI suitability assessment · Non-AI alternative · Use-case card · Value-risk-feasibility matrix · Go/no-go and kill criteria
Current-state / brownfield assessment	Explicit	P2 - Discover process and architecture Brownfield repo assessment P2 - Discover process and architecture Brownfield assessment
Baseline KPI definition — Before AI FDE intervention	Explicit	P3 - Frame problem, root cause and value Baseline KPIs P3 - Frame problem, root cause and value Baseline dataset · KPI tree
Process discovery and process mining	Supported	P2 - Discover process and architecture Process Discovery · Lean · Waste Elimination · DMAIC Define/Measure P2 - Discover process and architecture Process/value-stream maps
Root-cause analysis	Explicit	P3 - Frame problem, root cause and value Root-cause analysis

Capability	Mapping	Exact Operating Model source anchor(s)
No-AI / conventional-automation alternative assessment	Explicit	P4 - Triage regulation and qualify use case Non-AI alternative
AI intervention justification	Explicit	P4 - Triage regulation and qualify use case AI suitability assessment · Non-AI alternative · Use-case card · Value-risk-feasibility matrix · Go/no-go and kill criteria
Product judgement	Supported	P4 - Triage regulation and qualify use case Value-risk-feasibility matrix · Go/no-go and kill criteria P8 - Generate, test and select options Weighted trade-off matrix · Build/buy assessment · Provider comparison · Preliminary ADRs · Selected solution P19 - Prove value and tell the decision story Decision recommendation
User journey and service-design thinking	Not explicit	<i>Not explicitly named in the AI FDE Operating Model.</i>
Business-value engineering	Supported	P3 - Frame problem, root cause and value KPI tree · CTQs · Value hypothesis · Counter-metrics · Success/failure criteria P19 - Prove value and tell the decision story Benefits-realisation report · Before/after KPIs · Cost-to-value analysis · Counter-metrics · Waste reduction · Unintended effects
Benefits hypothesis and value-tree design	Explicit	P3 - Frame problem, root cause and value KPI tree · CTQs · Value hypothesis
Process improvement / DMAIC	Explicit	P2 - Discover process and architecture DMAIC Define/Measure P3 - Frame problem, root cause and value DMAIC Analyse P19 - Prove value and tell the decision story DMAIC Improve P20 - Evaluate AIMS and decide lifecycle state DMAIC Control
Domain modelling / DDD	Explicit	P5 - Model the domain Domain-Driven Design · Ubiquitous Language · Bounded Contexts · Domain Events
Requirements engineering	Supported	P13 - Approve ADRs and delivery specification Delivery specification · NFRs · API/data contracts · Agent specifications · Evaluation cases · Telemetry/logging specification · Rollback requirements · Traceability matrix · Backlog

Capability	Mapping	Exact Operating Model source anchor(s)
Requirements-to-evidence traceability	Explicit	P13 - Approve ADRs and delivery specification Traceability P13 - Approve ADRs and delivery specification Traceability matrix P15 - Evaluate, attack and independently assure Evaluation harness · Golden datasets · Model/RAG results · Agent task-completion results
Constraint, assumption and dependency management	Supported	P2 - Discover process and architecture Dependencies P13 - Approve ADRs and delivery specification NFRs · API/data contracts · Agent specifications · Rollback requirements · Traceability matrix · Backlog
Data engineering	Supported	P9 - Design information architecture Target data architecture · Data contracts · Semantic model · Metadata/provenance design · Retrieval architecture P14 - Engineer Data pipelines
Data quality, contracts, lineage and provenance	Explicit	P6 - Qualify data and knowledge Data Discovery · Quality · Lineage · Provenance · Permissible use · Knowledge discovery P9 - Design information architecture Data contracts · Semantic model · Metadata/provenance design P13 - Approve ADRs and delivery specification API/data contracts
Knowledge engineering	Supported	P6 - Qualify data and knowledge Knowledge discovery P6 - Qualify data and knowledge Data/knowledge inventories P9 - Design information architecture Semantic Layers · Ontology · Knowledge Graphs · Graph/Vector Databases
Taxonomy, ontology and semantic modelling	Explicit	P9 - Design information architecture Semantic Layers · Ontology · Knowledge Graphs · Graph/Vector Databases P9 - Design information architecture Semantic model

Capability	Mapping	Exact Operating Model source anchor(s)
Knowledge graph justification and engineering	Supported	P8 - Generate, test and select options Weighted trade-off matrix P9 - Design information architecture Knowledge Graphs · Graph/Vector Databases P9 - Design information architecture Conditional ontology and graph/vector schemas
Context engineering	Explicit	P10 - Design AI and application architecture Prompt/context design P14 - Engineer Prompt/configuration registry
Enterprise integration	Explicit	P10 - Design AI and application architecture Integration/deployment patterns P14 - Engineer Brownfield integration
API, event and data-contract engineering	Explicit	P5 - Model the domain Domain Events P9 - Design information architecture Data contracts P10 - Design AI and application architecture API contracts P13 - Approve ADRs and delivery specification API/data contracts
Brownfield modernization	Supported	P2 - Discover process and architecture Brownfield repo assessment P14 - Engineer Brownfield integration
Migration, coexistence, cutover and rollback strategy	Supported	P13 - Approve ADRs and delivery specification Rollback requirements P17 - Deploy progressively and integrate adoption Rollback decision

Capability	Mapping	Exact Operating Model source anchor(s)
Architecture	Explicit	P2 - Discover process and architecture Architecture Discovery P8 - Generate, test and select options Reference Architectures P9 - Design information architecture Target data architecture P10 - Design AI and application architecture Target C4 Context/Container/Component views · AI/RAG architecture P11 - Design agentic and multi-agent orchestration Approved, bounded and testable agentic architecture P12 - Design security, guardrails and supplier controls Guardrail architecture
Architecture Decision Records — ADRs	Explicit	P8 - Generate, test and select options Preliminary ADRs P9 - Design information architecture Data ADRs P10 - Design AI and application architecture Architecture ADRs P11 - Design agentic and multi-agent orchestration Autonomy-level ADR P13 - Approve ADRs and delivery specification Final ADR register
Software engineering	Supported	P14 - Engineer Source/laC · Prompt/configuration registry · Model/tool versions · Data pipelines · Agent definitions · Orchestration workflows · Unit/integration tests · CI/CD evidence · As-built C4 views
AI-assisted engineering / Cursor-style engineering discipline	Explicit	P14 - Engineer AI-Assisted Engineering · Spec-Driven Development using Cursor
AI-system design	Explicit	P10 - Design AI and application architecture Design AI and application architecture
Model-selection and model-substitution engineering	Supported	P8 - Generate, test and select options PoC/model/RAG results · Weighted trade-off matrix · Build/buy assessment · Provider comparison P10 - Design AI and application architecture Model-routing design
Prompt engineering and prompt lifecycle management	Supported	P10 - Design AI and application architecture Prompt/context design P14 - Engineer Prompt/configuration registry

Capability	Mapping	Exact Operating Model source anchor(s)
RAG / grounding engineering	Explicit	P8 - Generate, test and select options PoC/model/RAG results P10 - Design AI and application architecture AI/RAG architecture P15 - Evaluate, attack and independently assure Model/RAG results
Agentic-system engineering	Explicit	P11 - Design agentic and multi-agent orchestration Agentic AI · Single-agent versus multi-agent decision · Multi-Agent Orchestration · Human-in-the-loop · Agent communication and control P14 - Engineer Agent/workflow implementation P15 - Evaluate, attack and independently assure Model/RAG/agent/multi-agent Evals
Tool-contract and bounded-agency engineering	Explicit	P11 - Design agentic and multi-agent orchestration Tool/action catalogue · Identity/permission matrix P11 - Design agentic and multi-agent orchestration Termination/loop controls · Human approval/override/escalation matrix P17 - Deploy progressively and integrate adoption Bounded agent autonomy
Human-in-the-loop / Human-on-the-loop design	Explicit	P11 - Design agentic and multi-agent orchestration Human-in-the-loop P11 - Design agentic and multi-agent orchestration Human approval/override/escalation matrix P7 - Define evaluations, impacts and risks Oversight and transparency requirements P17 - Deploy progressively and integrate adoption Human-AI workflow integration
Deterministic control boundaries	Supported	P11 - Design agentic and multi-agent orchestration Identity/permission matrix · Shared-memory design · Handoff protocol · Termination/loop controls · Human approval/override/escalation matrix P12 - Design security, guardrails and supplier controls Guardrail architecture · Control matrix P13 - Approve ADRs and delivery specification Agent specifications
Security-by-Design	Supported	P12 - Design security, guardrails and supplier controls Design security, guardrails and supplier controls

Capability	Mapping	Exact Operating Model source anchor(s)
AI / agentic security	Explicit	P12 - Design security, guardrails and supplier controls AI Security P12 - Design security, guardrails and supplier controls Agentic attack-surface map P15 - Evaluate, attack and independently assure OWASP testing · Red teaming · Guardrail validation
Zero Trust authorization	Supported	P11 - Design agentic and multi-agent orchestration Identity/permission matrix P12 - Design security, guardrails and supplier controls Trust boundaries
Software and AI supply-chain security	Explicit	P12 - Design security, guardrails and supplier controls Supply-chain assurance P12 - Design security, guardrails and supplier controls Component register · SBOM/AIBOM · Exit plan
Privacy-by-Design	Not explicit	<i>Not explicitly named in the AI FDE Operating Model.</i>
Responsible AI	Explicit	P1 - Mandate and field immersion Responsible AI context P4 - Triage regulation and qualify use case Responsible AI P7 - Define evaluations, impacts and risks Responsible AI P15 - Evaluate, attack and independently assure Responsible AI testing
Human-factors and automation-bias assessment	Not explicit	<i>Not explicitly named in the AI FDE Operating Model.</i>
Governance	Explicit	P1 - Mandate and field immersion Governance RACI P13 - Approve ADRs and delivery specification Approve ADRs and delivery specification P16 - Prepare operations, recovery and regulatory evidence Operational RACI P20 - Evaluate AIMS and decide lifecycle state Internal audit · Management review
AI assurance case	Supported	P15 - Evaluate, attack and independently assure Assurance report · Residual-risk acceptance P16 - Prepare operations, recovery and regulatory evidence AI-system record · Technical documentation · Transparency and oversight records

Capability	Mapping	Exact Operating Model source anchor(s)
Regulatory and standards applicability analysis	Explicit	P4 - Triage regulation and qualify use case ISO/IEC 42001 · EU AI Act classification P7 - Define evaluations, impacts and risks ISO/IEC 42005 · ISO/IEC 42001 risk planning · EU AI Act requirements P16 - Prepare operations, recovery and regulatory evidence ISO/IEC 42001/42005 · EU AI Act P20 - Evaluate AIMS and decide lifecycle state ISO/IEC 42001 Check/Act
Legal, contractual, licensing and IP assessment	Not explicit	Not explicitly named in the AI FDE Operating Model.
Evaluation and TEVV	Explicit	P7 - Define evaluations, impacts and risks Evals P15 - Evaluate, attack and independently assure Model/RAG/agent/multi-agent Evals P18 - Monitor and validate operational resilience Continuous Evals
Golden, edge, adversarial and failure datasets	Supported	P7 - Define evaluations, impacts and risks Golden-set specification · Scenarios P15 - Evaluate, attack and independently assure Golden datasets P15 - Evaluate, attack and independently assure Red teaming P10 - Design AI and application architecture Failure-mode design
AI red teaming	Explicit	P15 - Evaluate, attack and independently assure Red teaming
Business-outcome evaluation	Explicit	P19 - Prove value and tell the decision story KPIs before/after P19 - Prove value and tell the decision story Benefits-realisation report · Before/after KPIs · Cost-to-value analysis
Token efficiency	Not explicit	Not explicitly named in the AI FDE Operating Model.
Token economics	Not explicit	Not explicitly named in the AI FDE Operating Model.
AI FinOps	Supported	P18 - Monitor and validate operational resilience Cost/availability reports P19 - Prove value and tell the decision story Cost-to-value analysis

Capability	Mapping	Exact Operating Model source anchor(s)
Total Cost of Ownership — TCO	Supported	P8 - Generate, test and select options Build/buy assessment · Provider comparison P19 - Prove value and tell the decision story Cost-to-value analysis
Cost-per-successful-business-outcome measurement	Supported	P19 - Prove value and tell the decision story Before/after KPIs · Cost-to-value analysis
Performance and latency engineering	Supported	P13 - Approve ADRs and delivery specification NFRs P18 - Monitor and validate operational resilience Operational performance, risk and resilience evidence
Scalability and capacity engineering	Supported	P13 - Approve ADRs and delivery specification NFRs P19 - Prove value and tell the decision story Evidence-backed scale/change/stop package
Reliability and resilience	Explicit	P10 - Design AI and application architecture Failure-mode design P16 - Prepare operations, recovery and regulatory evidence Recovery evidence P18 - Monitor and validate operational resilience Monitor and validate operational resilience
Graceful degradation and AI-disabled operation	Supported	P10 - Design AI and application architecture Failure-mode design P13 - Approve ADRs and delivery specification Rollback requirements P16 - Prepare operations, recovery and regulatory evidence Incident/rollback plans P17 - Deploy progressively and integrate adoption Rollback decision
Observability	Explicit	P13 - Approve ADRs and delivery specification Telemetry/Observability design P16 - Prepare operations, recovery and regulatory evidence Telemetry · Observability P18 - Monitor and validate operational resilience Telemetry · Observability

Capability	Mapping	Exact Operating Model source anchor(s)
AI / agent observability and traceability	Explicit	P13 - Approve ADRs and delivery specification Traceability P16 - Prepare operations, recovery and regulatory evidence Agent/LLM telemetry P18 - Monitor and validate operational resilience Model/tool/agent traces · Inter-agent interaction logs
Service management	Supported	P16 - Prepare operations, recovery and regulatory evidence Dashboards/alerts · Runbooks · Incident/rollback plans · SOPs/training P18 - Monitor and validate operational resilience Incident records · Updated runbooks
SLO / SLA / error-budget engineering	Not explicit	<i>Not explicitly named in the AI FDE Operating Model.</i>
Incident response	Explicit	P16 - Prepare operations, recovery and regulatory evidence Incident/rollback plans P18 - Monitor and validate operational resilience Incident records
AI-specific incident response	Supported	P16 - Prepare operations, recovery and regulatory evidence Agent/LLM telemetry · Dashboards/alerts · Runbooks · Incident/rollback plans P18 - Monitor and validate operational resilience Agent-loop alerts · Incident records
Business continuity and disaster recovery	Explicit	P16 - Prepare operations, recovery and regulatory evidence Pre-production Chaos Drill P16 - Prepare operations, recovery and regulatory evidence Recovery evidence P18 - Monitor and validate operational resilience Controlled Chaos Drills P18 - Monitor and validate operational resilience Chaos/recovery results
Change management	Explicit	P16 - Prepare operations, recovery and regulatory evidence Change management P20 - Evaluate AIMS and decide lifecycle state Change reassessment
User adoption and behavioural change	Explicit	P17 - Deploy progressively and integrate adoption Human–AI workflow integration P17 - Deploy progressively and integrate adoption Adoption dashboard · Override/workaround logs · Updated SOPs
Training and enablement	Explicit	P16 - Prepare operations, recovery and regulatory evidence SOPs/training

Capability	Mapping	Exact Operating Model source anchor(s)
Commercial judgement	Supported	P8 - Generate, test and select options Build/buy assessment · Provider comparison P12 - Design security, guardrails and supplier controls Supplier/model assessment P19 - Prove value and tell the decision story Cost-to-value analysis
Build-vs-buy-vs-partner analysis	Supported	P8 - Generate, test and select options Build/buy assessment · Provider comparison
Vendor management	Supported	P8 - Generate, test and select options Provider comparison P12 - Design security, guardrails and supplier controls Supplier/model assessment
Third-party risk management	Supported	P12 - Design security, guardrails and supplier controls Supplier/model assessment · Model/system cards · Component register · SBOM/AIBOM · Exit plan
Vendor concentration and exit strategy	Supported	P8 - Generate, test and select options Provider comparison P12 - Design security, guardrails and supplier controls Exit plan
Operating-model design	Supported	P1 - Mandate and field immersion Governance RACI P16 - Prepare operations, recovery and regulatory evidence Operational RACI P20 - Evaluate AIMS and decide lifecycle state Management-review decision · Control plan
Roles, RACI and decision-rights design	Explicit	P1 - Mandate and field immersion Governance RACI P11 - Design agentic and multi-agent orchestration Identity/permission matrix P11 - Design agentic and multi-agent orchestration Human approval/override/escalation matrix P16 - Prepare operations, recovery and regulatory evidence Operational RACI
Operational ownership	Supported	P1 - Mandate and field immersion Sponsor/owner P16 - Prepare operations, recovery and regulatory evidence Operational RACI

Capability	Mapping	Exact Operating Model source anchor(s)
Operational handover	Supported	P16 - Prepare operations, recovery and regulatory evidence Runbooks · Incident/rollback plans · SOPs/training · Recovery evidence P16 - Prepare operations, recovery and regulatory evidence Operationally and legally ready release P17 - Deploy progressively and integrate adoption Controlled live service and adoption evidence
Knowledge transfer	Supported	P16 - Prepare operations, recovery and regulatory evidence SOPs/training P21 - Retire and capture reusable IP Reusable ADRs/C4 templates · Updated reference architectures · Eval and guardrail templates
Production-readiness assessment	Supported	P15 - Evaluate, attack and independently assure Independently assured release candidate P16 - Prepare operations, recovery and regulatory evidence Operationally and legally ready release
Deployment and release governance	Explicit	P17 - Deploy progressively and integrate adoption Release manifest · Deployment record · Pilot/canary results · Autonomy-expansion record · Rollback decision
Benefits-realisation measurement	Explicit	P19 - Prove value and tell the decision story Benefits-realisation report
Post-intervention KPI measurement — After AI FDE intervention	Explicit	P19 - Prove value and tell the decision story Before/after KPIs
Before-vs-after KPI variance analysis	Explicit	P19 - Prove value and tell the decision story Before/after KPIs
Value leakage analysis	Supported	P19 - Prove value and tell the decision story Counter-metrics · Waste reduction · Unintended effects
Continuous improvement	Explicit	P19 - Prove value and tell the decision story DMAIC Improve · Lean P20 - Evaluate AIMS and decide lifecycle state DMAIC Control P21 - Retire and capture reusable IP Lessons learned · Reference Architecture improvement
Model / system drift management	Explicit	P18 - Monitor and validate operational resilience Drift P18 - Monitor and validate operational resilience Drift reports

Capability	Mapping	Exact Operating Model source anchor(s)
Retirement and decommissioning strategy	Explicit	P21 - Retire and capture reusable IP Safe retirement P21 - Retire and capture reusable IP Retirement plan
90-day / medium-term transformation roadmap	Not explicit	Not explicitly named in the AI FDE Operating Model.
Executive communication	Explicit	P19 - Prove value and tell the decision story Data-Driven Storytelling P19 - Prove value and tell the decision story Executive SCQA paper · C4 executive view · Decision recommendation
Elevator pitch and final defence	Supported	P19 - Prove value and tell the decision story Prove value and tell the decision story P19 - Prove value and tell the decision story Executive SCQA paper · C4 executive view · Decision recommendation

Legacy repo analysis

P2 - Discover process and architecture

Brownfield repo assessment

IMPERFECTION INCONSISTENCY FRICTION COMPLEXITY VOLATILITY UNCERTAINTY HIDDEN DEPENDENCY UNKNOWN UNKNOWN

Lens	What participants investigate
Imperfection	What is technically or operationally wrong?
Inconsistency	Which sources, states or rules contradict each other?
Friction	Where is unnecessary human/manual effort occurring?
Complexity	What is inherently difficult even when functioning correctly?
Volatility	What keeps changing dynamically?
Uncertainty	What cannot be known with complete confidence?
Hidden Dependency	What downstream consequence is not immediately visible?
Unknown Unknown	What has the existing enterprise never modelled or measured?

AI FDE

Operating Model



21 stages

Methods and disciplines • Essential artifacts • Output/input to next phase

SURYA

AI FDE Operating Model

Seq. • Workflow • Methods and disciplines • Essential artifacts • Output/input to next phase

01 Mandate and field immersion

METHODS AND DISCIPLINES

FDE ownership • Stakeholder discovery • Responsible AI context • ISO/IEC 42001 scope

ESSENTIAL ARTIFACTS

Engagement charter • Scope • Outcome statement • Sponsor/owner • Governance RACI • Stakeholder and affected-groups maps • Field-evidence register

OUTPUT/INPUT TO NEXT PHASE

Approved mandate and operating context

02 Discover process and architecture

METHODS AND DISCIPLINES

Process Discovery • Lean • Waste Elimination • DMAIC Define/Measure • Architecture Discovery • Brownfield repo assessment • C4

ESSENTIAL ARTIFACTS

SIPOC • Process/value-stream maps • Waste register • System landscape • Brownfield assessment • Current-state C4 views • Dependencies • Data flows • Trust boundaries

OUTPUT/INPUT TO NEXT PHASE

Current-state process and architecture baseline

03 Frame problem, root cause and value

METHODS AND DISCIPLINES

SCQA · DMAIC Analyse · Root-cause analysis · Baseline KPIs · Data-driven framing

ESSENTIAL ARTIFACTS

SCQA problem frame · Root-cause analysis · Baseline dataset · KPI tree · CTQs · Value hypothesis · Counter-metrics · Success/failure criteria

OUTPUT/INPUT TO NEXT PHASE

Evidence-backed problem and baseline

04 Triage regulation and qualify use case

METHODS AND DISCIPLINES

Responsible AI · ISO/IEC 42001 · EU AI Act classification · AI suitability · Solution Decoding

ESSENTIAL ARTIFACTS

Impact/regulatory screen · Prohibited-use check · AI suitability assessment · Non-AI alternative · Use-case card · Value-risk-feasibility matrix · Go/no-go and kill criteria

OUTPUT/INPUT TO NEXT PHASE

Approved and justified use case

05

Model the domain

METHODS AND DISCIPLINES

Domain-Driven Design · Ubiquitous Language · Bounded Contexts · Domain Events

ESSENTIAL ARTIFACTS

Glossary · Domain capability map · Business rules · Decision model · Domain events · Ownership map · DDD Context Map · Bounded Contexts

OUTPUT/INPUT TO NEXT PHASE

Domain and decision model

06

Qualify data and knowledge

METHODS AND DISCIPLINES

Data Discovery · Quality · Lineage · Provenance · Permissible use · Knowledge discovery

ESSENTIAL ARTIFACTS

Data/knowledge inventories · Lineage · Quality profile · Provenance · Access matrix · Representativeness assessment · Data-gap register · Dataset datasheets

OUTPUT/INPUT TO NEXT PHASE

Data and knowledge readiness assessment

07 Define evaluations, impacts and risks

METHODS AND DISCIPLINES

Evals · Responsible AI · ISO/IEC 42005 · ISO/IEC 42001 risk planning · EU AI Act requirements

ESSENTIAL ARTIFACTS

Evaluation strategy · Golden-set specification · Scenarios · Acceptance thresholds · AI impact assessment · Risk/harms registers · Risk treatment · Oversight and transparency requirements

OUTPUT/INPUT TO NEXT PHASE

Evaluation, impact and risk requirements

08 Generate, test and select options

METHODS AND DISCIPLINES

Solution Decoding · Reference Architectures · GenAI patterns · PoCs · Trade-off Matrices

ESSENTIAL ARTIFACTS

Solution catalogue · Reference-architecture comparison · PoC/model/RAG results · Weighted trade-off matrix · Build/buy assessment · Provider comparison · Preliminary ADRs · Selected solution

OUTPUT/INPUT TO NEXT PHASE

Approved solution and trade-offs

09

Design information architecture

METHODS AND DISCIPLINES

Semantic Layers · Ontology · Knowledge Graphs · Graph/Vector Databases · Data Architecture Decisions

ESSENTIAL ARTIFACTS

Target data architecture · Data contracts · Semantic model · Metadata/provenance design · Retrieval architecture · Data ADRs · Conditional ontology and graph/vector schemas

OUTPUT/INPUT TO NEXT PHASE

Approved information architecture

10

Design AI and application architecture

METHODS AND DISCIPLINES

C4 · GenAI patterns · RAG · Model routing · Prompt/context design · Integration/deployment patterns

ESSENTIAL ARTIFACTS

Target C4 Context/Container/Component views · AI/RAG architecture · Model-routing design · API contracts · Prompt/context design · Deployment topology · Failure-mode design · Architecture ADRs

OUTPUT/INPUT TO NEXT PHASE

Complete base AI/application architecture

11

Design agentic and multi-agent orchestration

METHODS AND DISCIPLINES

Agentic AI · Single-agent versus multi-agent decision · Multi-Agent Orchestration · Human-in-the-loop · Agent communication and control

ESSENTIAL ARTIFACTS

Agent-suitability assessment · Autonomy-level ADR · Agent responsibility map · Orchestration topology · Agent interaction/sequence diagram · State-machine model · Tool/action catalogue · Identity/permission matrix · Shared-memory design · Handoff protocol · Termination/loop controls · Human approval/override/escalation matrix

OUTPUT/INPUT TO NEXT PHASE

Approved, bounded and testable agentic architecture

12

Design security, guardrails and supplier controls

METHODS AND DISCIPLINES

AI Security · Threat Modelling · OWASP GenAI LLM Top 10 2026 · OWASP Agentic Top 10 2026 · Guardrails · Supply-chain assurance

ESSENTIAL ARTIFACTS

Threat model · Agentic attack-surface map · Trust boundaries · Abuse cases · OWASP mapping · Guardrail architecture · Control matrix · Supplier/model assessment · Model/system cards · Component register · SBOM/AIBOM · Exit plan

OUTPUT/INPUT TO NEXT PHASE

Controlled architecture and approved components

13

Approve ADRs and delivery specification

METHODS AND DISCIPLINES

ADRs · Spec-Driven Development · Telemetry/Observability design · Traceability

ESSENTIAL ARTIFACTS

Final ADR register · Approved C4 baseline · Delivery specification · NFRs · API/data contracts · Agent specifications · Evaluation cases · Telemetry/logging specification · Rollback requirements · Traceability matrix · Backlog

OUTPUT/INPUT TO NEXT PHASE

Build-ready specification

14

Engineer

METHODS AND DISCIPLINES

AI-Assisted Engineering · Spec-Driven Development using Cursor · Brownfield integration · Agent/workflow implementation

ESSENTIAL ARTIFACTS

Source/laC · Prompt/configuration registry · Model/tool versions · Data pipelines · Agent definitions · Orchestration workflows · Unit/integration tests · CI/CD evidence · As-built C4 views

OUTPUT/INPUT TO NEXT PHASE

Versioned deployable increments

15

Evaluate, attack and independently assure

METHODS AND DISCIPLINES

Model/RAG/agent/multi-agent Evals · Responsible AI testing · OWASP testing · Red teaming · Guardrail validation

ESSENTIAL ARTIFACTS

Evaluation harness · Golden datasets · Model/RAG results · Agent task-completion results · Inter-agent handoff tests · Coordination/conflict tests · Loop/termination tests · Prompt-injection/tool-misuse tests · Assurance report · Residual-risk acceptance

OUTPUT/INPUT TO NEXT PHASE

Independently assured release candidate

16

Prepare operations, recovery and regulatory evidence

METHODS AND DISCIPLINES

Telemetry · Observability · Change management · Pre-production Chaos Drill · ISO/IEC 42001/42005 · EU AI Act

ESSENTIAL ARTIFACTS

Operational RACI · Agent/LLM telemetry · Dashboards/alerts · Runbooks · Incident/rollback plans · SOPs/training · Recovery evidence · AI-system record · Technical documentation · Transparency and oversight records

OUTPUT/INPUT TO NEXT PHASE

Operationally and legally ready release

17

Deploy progressively and integrate adoption

METHODS AND DISCIPLINES

Shadow/pilot/canary · Bounded agent autonomy · Human–AI workflow integration

ESSENTIAL ARTIFACTS

Release manifest · Deployment record · Pilot/canary results · Autonomy-expansion record · Rollback decision · Adoption dashboard · Override/workaround logs · Updated SOPs

OUTPUT/INPUT TO NEXT PHASE

Controlled live service and adoption evidence

18

Monitor and validate operational resilience

METHODS AND DISCIPLINES

Telemetry · Observability · Continuous Evals · Agent monitoring · Drift · Controlled Chaos Drills

ESSENTIAL ARTIFACTS

Operational dashboards · Model/tool/agent traces · Inter-agent interaction logs · Drift reports · Cost/availability reports · Incident records · Agent-loop alerts · Chaos/recovery results · Updated runbooks

OUTPUT/INPUT TO NEXT PHASE

Operational performance, risk and resilience evidence

19

Prove value and tell the decision story

METHODS AND DISCIPLINES

KPIs before/after · DMAIC Improve · Lean · SCQA · Data-Driven Storytelling

ESSENTIAL ARTIFACTS

Benefits-realisation report · Before/after KPIs · Cost-to-value analysis · Counter-metrics · Waste reduction · Unintended effects · Executive SCQA paper · C4 executive view · Decision recommendation

OUTPUT/INPUT TO NEXT PHASE

Evidence-backed scale/change/stop package

20

Evaluate AIMS and decide lifecycle state

METHODS AND DISCIPLINES

ISO/IEC 42001 Check/Act · Internal audit · Management review · DMAIC Control · Change reassessment

ESSENTIAL ARTIFACTS

AIMS performance report · Audit findings · CAPA register · Management-review decision · Control plan · Updated impact/risk/C4/ADRs/evaluations · Scale/change/restrict/suspend/retire decision

OUTPUT/INPUT TO NEXT PHASE

Authorised lifecycle state or return to an earlier phase

21

Retire and capture reusable IP

METHODS AND DISCIPLINES

Safe retirement · Record control · Lessons learned · Reference Architecture improvement

ESSENTIAL ARTIFACTS

Retirement plan · Notifications · Access revocation · Data/model/memory disposition · Agent/tool credential revocation · Supplier/dependency closure · Reusable ADRs/C4 templates · Updated reference architectures · Eval and guardrail templates

OUTPUT/INPUT TO NEXT PHASE

Formally closed system and reusable FDE IP